

1 Complex manifolds

Definition 1.1. A *complex manifold* of dimension n consists of the following:

- (1) a Hausdorff, second countable topological space M ;
- (2) an open cover

$$M = \bigcup_{\alpha \in \Lambda} U_\alpha$$

- (3) for each $\alpha \in \Lambda$ a homeomorphism

$$\varphi_\alpha : U_\alpha \rightarrow \tilde{U}_\alpha$$

where $\tilde{U}_\alpha$ is an open subset of $\mathbb{C}^n$ such that the transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1}$$

are *holomorphic* in their domain of definition.

In this case, we say that $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ is a *holomorphic atlas* for M . Two holomorphic atlases are equivalent if their union is also a holomorphic atlas.

A complex manifold is a differentiable manifold equipped with an equivalence class of holomorphic atlases.

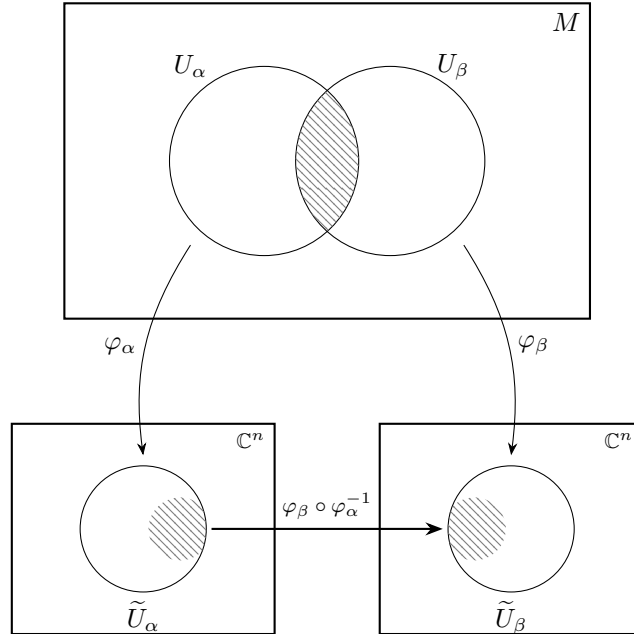


Figure 1: A complex manifold and transition maps.

An element (U, φ) of an holomorphic atlas in the equivalence class defining M is called a *chart*. In practice, we write z instead of φ and refer to $z = (z_1, \dots, z_n)$ as *local complex coordinates*.

Example 1.2. An open subset U of $\mathbb{C}^n$ is a complex manifold defined by an atlas consisting of the single chart (U, φ) with φ equal to the identity map.

Example 1.3 (The Riemann sphere). Let $S^2 \subset \mathbb{R}^3$ be the 2-sphere

$$S^2 = \{(x, y, t) \in \mathbb{R}^3 \mid x^2 + y^2 + t^2 = 1\}.$$

Let $U_N = S^2 \setminus \{N\}$ and $U_S = S^2 \setminus \{S\}$, where $N = (0, 0, 1)$ and $S = -N$ are the north and south poles. Let σ_N and σ_S be the corresponding stereographic projections to the xy -plane, $\sigma_N : U_N \rightarrow \mathbb{R}^2$ and $\sigma_S : U_S \rightarrow \mathbb{R}^2$, given by

$$\sigma_N(x, y, t) = \frac{1}{1-t} \cdot (x, y) \quad \text{and} \quad \sigma_S(x, y, t) = \frac{1}{1+t} \cdot (x, y)$$

The above equations follow after noticing the similar triangles in the figure: $\triangle TNP \sim \triangle ONP'$ and $\triangle TSP \sim \triangle OSP''$.

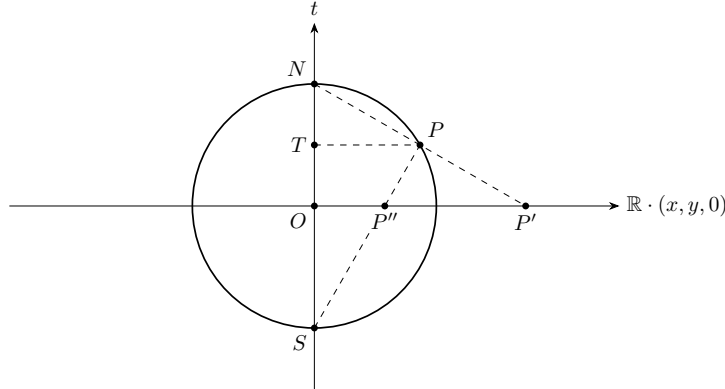


Figure 2: Here $P' = \sigma_N(P)$ and $P'' = \sigma_S(P)$ are the stereographic projections from the north and south poles of the point $P = (x, y, t)$ lying on the 2-sphere, and $T = (0, 0, t)$ is the projection of P to the t -axis.

Using the standard identification of $\mathbb{R}^2$ with $\mathbb{C}$ via $(x, y) \mapsto x + iy$, we can regard σ_N and σ_S as homeomorphisms $\sigma_N : U_N \rightarrow \mathbb{C}$ and $\sigma_S : U_S \rightarrow \mathbb{C}$. Define $\varphi_N = \sigma_N$ and $\varphi_S = \bar{\sigma}_S$ (complex conjugate). Let (x, y, t) be a point in $U_N \cap U_S$ and let $z = \varphi_N(x, y, t)$ and $w = \varphi_S(x, y, t)$. We calculate zw (product of complex numbers) as follows:

$$zw = \frac{1}{(1-t^2)}(x+iy)(x-iy) = 1$$

where in the second equality we used that $x^2 + y^2 + t^2 = 1$. Therefore

$$(\varphi_S \circ \varphi_N^{-1})(z) = \frac{1}{z}$$

is holomorphic on its domain of definition $\varphi_N(U_N \cap U_S) = \mathbb{C} \setminus \{0\}$. We conclude that $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ with $\Lambda = \{N, S\}$ is a holomorphic atlas for S^2 . This atlas endows the 2-sphere with the structure of a complex manifold of dimension 1, i.e., a *Riemann surface*.

An equivalent complex manifold is obtained as follows. Let $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the extended complex plane endowed with the one-point compactification topology, i.e., $U \subset \widehat{\mathbb{C}}$ is open if either $U \subset \mathbb{C}$ is open or $U = \widehat{\mathbb{C}} \setminus K$ with $K \subset \mathbb{C}$ compact. There is a natural homeomorphism

$$F : \widehat{\mathbb{C}} \rightarrow S^2, \quad F(z) = \begin{cases} \sigma_N^{-1}(z) & \text{if } z \in \mathbb{C} \\ N & \text{if } z = \infty. \end{cases}$$

We use F to pull-back the holomorphic atlas we constructed on S^2 . This gives us an holomorphic atlas $\{(U_1, \varphi_1), (U_2, \varphi_2)\}$ for $\widehat{\mathbb{C}}$ with

$$\varphi_1 : \mathbb{C} \rightarrow \mathbb{C}, \quad \varphi_1(z) = z, \quad \varphi_2 : \widehat{\mathbb{C}} \setminus \{0\} \rightarrow \mathbb{C}, \quad \varphi_2(z) = 1/z$$

(here $1/\infty = 0$). The complex manifold $\widehat{\mathbb{C}}$ is known as the *Riemann sphere*.

Example 1.4 (Products). The product of two complex manifolds is a complex manifold, in an obvious way. For instance, $\widehat{\mathbb{C}} \times \widehat{\mathbb{C}}$ is a complex manifold of dimension 2 (a *complex surface*) homeomorphic to $S^2 \times S^2$.

Example 1.5 (Complex projective space). As a set, complex projective space $\mathbb{CP}^n$ is the set of all 1-dimensional complex vector subspaces of $\mathbb{C}^{n+1}$. As a topological space, it is endowed with the quotient topology induced by the map

$$\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n, \quad (Z_0, \dots, Z_n) \rightarrow [Z_0, \dots, Z_n]$$

which sends a non-zero vector to the line it spans. The *standard open covering* of $\mathbb{CP}^n$ is given by the $n+1$ open subsets $U_0, \dots, U_n$ with

$$U_i = \{[Z_0, \dots, Z_n] \mid Z_i \neq 0\}.$$

Define homeomorphisms $\varphi_i : U_i \rightarrow \mathbb{C}^n$ given by

$$\varphi_i([Z_0, \dots, Z_n]) = \left(\frac{Z_0}{Z_i}, \dots, \frac{Z_{i-1}}{Z_i}, \frac{Z_{i+1}}{Z_i}, \dots, \frac{Z_n}{Z_i} \right).$$

More geometrically, $\varphi_i([Z_0, \dots, Z_n])$ is the point of intersection of the complex line $\mathbb{C} \cdot (Z_0, \dots, Z_n)$ with the affine subspace

$$A_i = \{Z \in \mathbb{C}^{n+1} \mid Z_i = 1\} \cong \mathbb{C}^n.$$

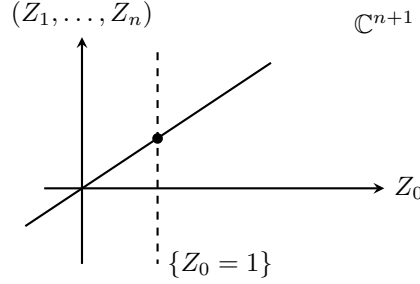


Figure 3: Affine chart U_0 on $\mathbb{CP}^n$ visualized as the hyperplane $Z_0 = 1$.

To write the transition maps, let $\tilde{U}_{i,j}$ be the open subset of A_i given by

$$\tilde{U}_{i,j} = \{Z \in A_i \mid Z_j \neq 0\}.$$

Then $\varphi_j \circ \varphi_i^{-1} : \tilde{U}_{i,j} \rightarrow \tilde{U}_{j,i}$ is the holomorphic map given by

$$(Z_0, \dots, Z_n) \mapsto Z_j^{-1} \cdot (Z_0, \dots, Z_n).$$

The holomorphic atlas $\{(U_i, \varphi_i) \mid i = 0, \dots, n\}$ endows $\mathbb{CP}^n$ with the structure of a complex manifold of dimension n .

Example 1.6 (Projective manifolds). A *projective variety* is the common zero set of a finite collection of homogeneous polynomials $f_i \in \mathbb{C}[Z_0, \dots, Z_n]$,

$$\bigcap_{i=1}^k \{f_i = 0\} \subset \mathbb{CP}^n.$$

A *projective manifold* is a smooth projective variety.

A *hypersurface* $M = \{f = 0\} \subset \mathbb{CP}^n$ is the zero set of a homogeneous polynomial $f \in \mathbb{C}[Z_0, \dots, Z_n]$ of degree $d \geq 1$. If $d = 1$ then, after a linear change of variables, we can assume that $f = Z_n$ and $M \cong \mathbb{CP}^{n-1}$. If $d \geq 2$ then the hypersurface M is smooth if and only if the only solution $(Z_0, \dots, Z_n)$ in $\mathbb{C}^{n+1}$ to the system of equations

$$\frac{\partial f}{\partial Z_i} = 0 \quad \text{for } i = 0, \dots, n$$

is the trivial solution $(Z_0, \dots, Z_n) = (0, \dots, 0)$; in which case M is a complex manifold of dimension $n - 1$.

For example, if $f \in \mathbb{C}[Z_0, \dots, Z_n]$ is the homogeneous polynomial

$$f = Z_0^d + \dots + Z_n^d$$

then the partial derivatives of f are

$$\frac{\partial f}{\partial Z_i} = dZ_i^{d-1}.$$

It is immediate that there is no point in $\mathbb{C}^{n+1} \setminus \{0\}$ where all the partial derivatives of f vanish simultaneously. Thus $\{f = 0\} \subset \mathbb{CP}^n$ is a smooth hypersurface, known as the Fermat hypersurface of degree d . If $n = 2$ then $\{f = 0\}$ is a Riemann surface of genus

$$g = \frac{(d-1)(d-2)}{2}.$$

Definition 1.7. Let M and N be two complex manifolds and let $f : M \rightarrow N$ be a continuous map. The map f is *holomorphic* if for any holomorphic charts (U, φ) and (V, ψ) of M and N , the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic on its domain of definition.

The complex manifolds M and N are *biholomorphic* if there is a holomorphic bijection $f : M \rightarrow N$ whose inverse $f^{-1} : N \rightarrow M$ is also holomorphic.

Example 1.8. The upper half space $H = \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$ and the unit disc $D = \{z \in \mathbb{C} \mid |z| < 1\}$ are biholomorphic via

$$z \mapsto \frac{z-i}{z+i}.$$

The *Riemann mapping theorem* asserts that if $U \subset \mathbb{C}$ is a proper simply connected open set then U is biholomorphic to the unit disc. The complex plane is not biholomorphic to the unit disc because any holomorphic map $f : \mathbb{C} \rightarrow D$ is constant by Liouville's theorem.

Example 1.9. The map $f : \widehat{\mathbb{C}} \rightarrow \mathbb{CP}^1$ given by

$$f(z) = \begin{cases} [z, 1] & \text{if } z \in \mathbb{C} \\ [1, 0] & \text{if } z = \infty \end{cases}$$

is a biholomorphism.

Example 1.10 (Hopf map). Restricting the projection $\mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$ to the unit sphere, we obtain a smooth submersion

$$H : S^{2n+1} \rightarrow \mathbb{CP}^n$$

whose fibre over $[Z_0, \dots, Z_n]$ is the circle $e^{i\theta} \cdot (Z_0, \dots, Z_n)$ for $e^{i\theta} \in S^1$. Identifying $\mathbb{CP}^1$ with S^2 via stereographic projection, we obtain a map $H : S^3 \rightarrow S^2$. To give a formula for this map, we compose $z = Z_0/Z_1$ with the inverse stereographic projection

$$\sigma_N^{-1}(z) = \frac{1}{|z|^2 + 1} (2z, |z|^2 - 1)$$

and use that $|Z_0|^2 + |Z_1|^2 = 1$ to obtain

$$H(Z_0, Z_1) = (2Z_0\bar{Z}_1, |Z_0|^2 - |Z_1|^2)$$

Example 1.11. The map $\mathbb{CP}^1 \rightarrow \mathbb{CP}^2$ given by

$$[Z_0, Z_1] \mapsto [Z_0^2, Z_0 Z_1, Z_1^2]$$

defines a holomorphic embedding whose image is the smooth plane curve

$$\{[Y_0, Y_1, Y_2] \mid Y_0 Y_2 = Y_1^2\} \subset \mathbb{CP}^2.$$

This map is known as the degree 2 Veronese embedding.

Example 1.12. The complex surface $\mathbb{CP}^1 \times \mathbb{CP}^1$ is biholomorphic to a smooth degree 2 hypersurface (a *quadric*) in $\mathbb{CP}^3$, as given by the Segre embedding

$$([X_0, X_1], [Y_0, Y_1]) \mapsto [X_0 Y_0, X_0 Y_1, X_1 Y_0, X_1 Y_1]$$

whose image is the smooth quadric

$$\{Z_0 Z_3 - Z_1 Z_2 = 0\} \subset \mathbb{CP}^3.$$

The following results illustrate some sharp contrasts between the smooth and holomorphic categories.

- If M is a compact connected complex manifold and $f : M \rightarrow \mathbb{C}$ is holomorphic then f is constant. This is a consequence of the maximum principle.
- If M is a compact complex manifold then its group of automorphisms $\text{Aut}(M)$ (i.e. biholomorphisms $f : M \rightarrow M$) is finite dimensional.

It is perhaps easier to understand the infinitesimal version of the second bullet: the set of all holomorphic vector fields on a compact complex manifold M is a finite dimensional vector space. The general fact is that if $E \rightarrow M$ is a holomorphic vector bundle then the set of global holomorphic sections of E is a finite dimensional vector space because the $\bar{\partial}$ operator is elliptic.

Example 1.13. The group $\text{Aut}(\mathbb{CP}^n)$ is isomorphic to the projective linear group $\text{PGL}(n+1, \mathbb{C})$. Under the isomorphism $\widehat{\mathbb{C}} \cong \mathbb{CP}^1$ it corresponds to the action by Möbius transformations:

$$z \mapsto \frac{az + b}{cz + d}.$$

Example 1.14. The *polydisc* D^2 is *not* biholomorphic to the unit ball $B \subset \mathbb{C}^2$. Indeed, $\text{Aut}(B) \cong \text{PU}(2, 1)$ is connected while $\text{Aut}(D^2) \cong \text{Aut}(D)^2 \rtimes S_2$ has two connected components, one containing the identity and the other containing the swap map $(z, w) \mapsto (w, z)$.

Example 1.15. The group $\text{Aut}(\mathbb{C})$ consists of affine maps $f(z) = az + b$ with $a, b \in \mathbb{C}$ and $a \neq 0$. But behold, if $n \geq 2$ then the group $\text{Aut}(\mathbb{C}^n)$ is infinite dimensional. Indeed, if f is any entire holomorphic function on $\mathbb{C}$ then the shear map

$$(z, w) \mapsto (z, w + f(z))$$

is a biholomorphism of $\mathbb{C}^2$. Somewhat ironically, the simplest complex manifold $\mathbb{C}^n$ has one of the most complicated automorphism groups.

2 Kähler metrics

Let M be a complex manifold of dimension n . Denote by TM the tangent bundle of the underlying smooth manifold, thus TM is a real vector bundle over M of rank $2n$.

Lemma 2.1. *There is a smooth section I of the bundle $\text{End}(TM)$ uniquely determined by the following condition: if $z = (z_1, \dots, z_n)$ are local holomorphic coordinates on M with $z_j = x_j + iy_j$ then*

$$I \frac{\partial}{\partial x_j} = \frac{\partial}{\partial y_j} \quad \text{and} \quad I \frac{\partial}{\partial y_j} = -\frac{\partial}{\partial x_j}$$

Proof. Let $w = (w_1, \dots, w_n)$ be another system of holomorphic coordinates with $w_j = u_j + iv_j$. Define I' by

$$I' \frac{\partial}{\partial u_j} = \frac{\partial}{\partial v_j} \quad \text{and} \quad I' \frac{\partial}{\partial v_j} = -\frac{\partial}{\partial u_j}.$$

We want to show that $I' = I$, thus it suffices to prove

$$I' \frac{\partial}{\partial x_j} = \frac{\partial}{\partial y_j} \quad \text{and} \quad I' \frac{\partial}{\partial y_j} = -\frac{\partial}{\partial x_j}.$$

Write

$$\frac{\partial}{\partial x_j} = \sum_{i=1}^n \frac{\partial u_i}{\partial x_j} \frac{\partial}{\partial u_i} + \sum_{i=1}^n \frac{\partial v_i}{\partial x_j} \frac{\partial}{\partial v_i}.$$

Applying I' we obtain

$$I' \frac{\partial}{\partial x_j} = \sum_{i=1}^n \frac{\partial u_i}{\partial x_j} \frac{\partial}{\partial v_i} - \sum_{i=1}^n \frac{\partial v_i}{\partial x_j} \frac{\partial}{\partial u_i}.$$

Since the coordinate functions $w_i = u_i + iv_i$ are holomorphic, they satisfy the Cauchy-Riemann equations:

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial v_i}{\partial y_j} \quad \text{and} \quad \frac{\partial u_i}{\partial y_j} = -\frac{\partial v_i}{\partial x_j}.$$

Thus

$$\begin{aligned} I' \frac{\partial}{\partial x_j} &= \sum_{i=1}^n \frac{\partial v_i}{\partial y_j} \frac{\partial}{\partial v_i} + \sum_{i=1}^n \frac{\partial u_i}{\partial y_j} \frac{\partial}{\partial u_i} \\ &= \frac{\partial}{\partial y_j}. \end{aligned}$$

Similarly, $I' \partial / \partial y_j = -\partial / \partial x_j$ (or alternatively use that both I and I' square to minus the identity). $\square$

The smooth section I of $\text{End}(TM)$ given by the above lemma satisfies

$$I^2 = -1.$$

We refer to I as the *complex structure* of M .

Definition 2.2. A Riemannian metric g on M is *Hermitian* if it satisfies

$$g(Iu, Iv) = g(u, v)$$

for every pair of tangent vectors $u, v \in T_p M$.

Any complex manifold admits Hermitian metrics: if g_0 is any Riemannian metric then $g(u, v) = g_0(u, v) + g_0(Iu, Iv)$ is a Hermitian metric.

Lemma 2.3. Let g be an Hermitian metric on M . Define

$$\omega(u, v) = g(Iu, v)$$

Then ω is a 2-form on M .

Proof. Being a 2-form means bilinear (which is immediate since I is linear and g bilinear) and anti-symmetric, which is easily checked:

$$\omega(v, u) = g(Iv, u) = g(I^2 v, Iu) = g(-v, Iu) = -g(Iu, v) = -\omega(u, v).$$

□

The 2-form ω is called the *fundamental 2-form* of g . The tangent bundle TM equipped with the complex structure I can be regarded as a complex vector bundle (of rank n) with scalar multiplication by complex numbers defined as

$$(a + ib)v = av + bIv$$

Then the complex valued real bilinear form $h : TM \times TM \rightarrow \mathbb{C}$ defined as

$$h(u, v) = g(u, v) - i\omega(u, v)$$

is a Hermitian inner product on TM , meaning that: it is $\mathbb{C}$ -linear on the first factor $h(\lambda u, v) = \lambda h(u, v)$ for $\lambda \in \mathbb{C}$; it is conjugate symmetric $h(v, u) = \overline{h(u, v)}$; and it is positive definite $h(v, v) > 0$ for all $v \neq 0$. Hence the name Hermitian metric for the Riemannian metric g .

Definition 2.4. Let M be a complex manifold and let g be a Hermitian metric on M . The metric g is *Kähler* if its fundamental 2-form ω is closed:

$$d\omega = 0$$

where d is the exterior derivative.

The fundamental 2-form ω of a Kähler metric is a *symplectic form* on M . Thus, Kähler manifolds are symplectic. If M is compact symplectic manifold then the second de Rham cohomology group $H_{dR}^2(M, \mathbb{R})$ must be non-zero. Not all complex manifolds satisfy this topological condition. For example the quotient of $\mathbb{C}^2 \setminus \{0\}$ by the $\mathbb{Z}$ -action generated by $(z, w) \mapsto 2 \cdot (z, w)$ is a complex manifold homeomorphic to $S^1 \times S^3$ which has $b_2 = 0$. Thus, this complex manifold does not admit any Kähler metrics at all.

Example 2.5. The Euclidean metric on $\mathbb{C}^n$ with coordinates $z_j = x_j + iy_j$

$$g = \sum_{j=1}^n (dx_j^2 + dy_j^2)$$

is Kähler. Its fundamental 2-form is the constant 2-form

$$\omega = \sum_{j=1}^n dx_j \wedge dy_j$$

Example 2.6 (Fubini-Study metric). Let $\pi : S^{2n+1} \rightarrow \mathbb{CP}^n$ be the Hopf fibration. Decompose the tangent bundle of the sphere as

$$TS^{2n+1} = V \oplus H$$

where V is tangent to the circle directions and $H = V^\perp$. Let $p \in \mathbb{CP}^n$ and choose $\tilde{p} \in S^{2n+1}$ with $\pi(\tilde{p}) = p$. Given a vector $v \in T_p \mathbb{CP}^n$ there is a unique $\tilde{v} \in H_{\tilde{p}}$ with $d\pi(\tilde{v}) = v$. The Fubini-Study metric g is defined as follows:

$$g(u, v) = g_0(\tilde{u}, \tilde{v})$$

where g_0 is the standard metric of the unit sphere S^{2n+1} . The fact that g_0 is invariant by the circle action implies that g is well defined, i.e., independent of the choice of point $\tilde{p}$.

The Fubini-Study metric is invariant under the action of the projective linear unitary group $PU(n+1)$. Proof: let $F \in PU(n+1)$ and let $\tilde{F} \in U(n+1)$ be a representative, the action of $PU(n+1)$ is defined as follows $F(p) = \pi(\tilde{F}(\tilde{p}))$ (since $\tilde{F}$ commutes with scalar multiplication the point $\pi(\tilde{F}(\tilde{p}))$ is independent of the choice of $\tilde{p}$) thus $d\pi \circ d\tilde{F} = dF \circ d\pi$. Since $\tilde{F}$ is unitary it preserves the orthogonal splitting $V \oplus H$ and the metric g_0 , thus

$$g(dF(u), dF(v)) = g_0(d\tilde{F}(\tilde{u}), d\tilde{F}(\tilde{v})) = g_0(\tilde{u}, \tilde{v}) = g(u, v).$$

The Fubini-Study metric is Hermitian. Proof: Suppose first that $p = [1, 0, \dots, 0]$. Let $\tilde{F}$ be the unitary transformation $\tilde{F} = \text{diag}(1, i, \dots, i)$ and let $F \in PU(n+1)$ the induced projective transformation. Then $F(p) = p$ and $dF(p) = I_p$ (the complex structure of $\mathbb{CP}^n$ at p), indeed in local coordinates $z_i = Z_i/Z_0$ for $i = 1, \dots, n$ the map F is scalar multiplication by i . Thus

$$g(Iu, Iv) = g(dF(u), dF(v)) = g(u, v).$$

For an arbitrary point $q \in \mathbb{CP}^n$ and vectors $u, v \in T_q \mathbb{CP}^n$ we argue as follows. First note that $U(n+1)$ acts transitively on S^{2n+1} because any unit vector can be completed to a unitary basis. In particular, we can always find $G \in PU(n+1)$ such that $G(p) = q$. The projective unitary transformation $H = G \circ F \circ G^{-1}$ fixes the point q and $dH(q) = I_q$ (to see this note that dG commutes with I because G is holomorphic). The same argument shows that $g(Iu, Iv) = g(u, v)$.

The Fubini-Study metric is Kähler. Let ω be the fundamental 2-form of g and let $\eta = d\omega$ be its exterior derivative. Thus η is a 3-form and we want to show that $\eta = 0$. Since exterior derivative commutes with pullback, the form η is invariant under $PU(n+1)$. Suppose that $p = [1, 0, \dots, 0]$. Let $\tilde{F}$ be the unitary transformation $\tilde{F} = \text{diag}(1, -1, \dots, -1)$ and let $F \in PU(n+1)$ the induced projective transformation, so $F(p) = p$ and $dF(p) = -1$. Then

$$\eta(u, v, w) = \eta(dF(u), dF(v), dF(w)) = \eta(-u, -v, -w) = -\eta(u, v, w).$$

We conclude that η vanishes at p , and since $PU(n+1)$ acts transitively on $\mathbb{CP}^n$, the form η must vanish everywhere. This finishes the proof that g is Kähler.

Alternatively, to prove that $d\omega = 0$. One can show that $\pi^*\omega = \omega_0|_{TS^{2n+1}}$ where ω_0 is the standard symplectic form of $\mathbb{R}^{2n+2}$. Thus $d\omega = 0$ follows from $d\omega_0 = 0$. This last point of view generalizes to the context of *symplectic reduction*.

Example 2.7. If $M \subset \mathbb{CP}^n$ is a complex projective manifold, then the restriction of the Fubini-Study metric to M is Kähler.

3 Calabi-Yau metrics

Theorem 3.1 (Yau, 1978). *Let (M, ω) be a compact Kähler manifold with $c_1(M) = 0$. Then there exists a unique Kähler metric ω_{CY} in the de Rham cohomology class of $[\omega] \in H^2(M, \mathbb{R})$ such that $\text{Ric}(\omega_{CY}) = 0$.*

For example, the theorem implies that the quartic K3 surface

$$M = \{[Z_0, \dots, Z_3] \in \mathbb{CP}^3 \mid Z_0^4 + \dots + Z_3^4 = 0\}$$

admits Calabi-Yau (= Kähler + Ricci-flat) metrics.

Ricci curvature and volume form. Let g be a Kähler metric on M . Fix a point $p \in M$. There exist holomorphic coordinates $(z^1, \dots, z^n)$ centered at p such that

$$g_{i\bar{j}}(p) = \delta_{ij}, \quad \partial_k g_{i\bar{j}}(p) = 0.$$

In these coordinates, the metric admits the expansion

$$g_{i\bar{j}}(z) = \delta_{ij} - \sum_{k,l} R_{i\bar{j}k\bar{l}}(p) z^k \bar{z}^l + O(|z|^3),$$

where $R_{i\bar{j}k\bar{l}}$ is the full Riemann curvature tensor.

The volume form of the metric is

$$\mathrm{dvol}_g = \det(g_{i\bar{j}}) dz^1 \wedge d\bar{z}^1 \cdots dz^n \wedge d\bar{z}^n.$$

Thus, the *volume density* is $\det(g_{i\bar{j}})$. Write

$$g = I + A(z), \quad A_{i\bar{j}} = -R_{i\bar{j}k\bar{l}}(p) z^k \bar{z}^l + O(|z|^3).$$

Using the standard expansion

$$\det(I + A) = 1 + \mathrm{tr}(A) + O(A^2),$$

we obtain $\det(g_{i\bar{j}}) = 1 + \mathrm{tr}(A) + O(|z|^4)$ with

$$\mathrm{tr}(A) = - \sum_{i,k,l} R_{i\bar{i}k\bar{l}}(p) z^k \bar{z}^l.$$

By definition of the Ricci tensor:

$$\sum_i R_{i\bar{i}k\bar{l}} = \mathrm{Ric}_{k\bar{l}}.$$

Hence,

$$\det(g_{i\bar{j}}) = 1 - \mathrm{Ric}_{k\bar{l}}(p) z^k \bar{z}^l + O(|z|^4).$$

Taking logarithms gives

$$\log \det(g_{i\bar{j}}) = -\mathrm{Ric}_{k\bar{l}}(p) z^k \bar{z}^l + O(|z|^4).$$

Taking two derivatives, we obtain:

$$\mathrm{Ric}_{k\bar{l}}(p) = \partial_k \partial_{\bar{l}} \log \det(g_{i\bar{j}}) \tag{1}$$

Monge-Ampère equation. The assumption that $c_1(M) = 0$ implies that there exists a volume form μ_{CY} on M such that, locally

$$\mu_{CY} = \Omega \wedge \bar{\Omega}$$

where Ω is a holomorphic volume form. Yau's theorem is proved by finding a smooth real function φ on M such that

$$(\omega + i\partial\bar{\partial}\varphi)^n = \mu_{CY} \tag{2}$$

If φ is such a solution, then $\omega_{CY} := \omega + i\partial\bar{\partial}\varphi$ has vanishing Ricci curvature. In local coordinates, Equation (2) implies that the volume density of ω_{CY} is $|f|^2$ where f is a nowhere vanishing holomorphic function. It follows from Equation (1) that the Ricci curvature of ω_{CY} is

$$\mathrm{Ric}_{k\bar{l}} = \partial_k \partial_{\bar{l}} \log |f|^2 = 0$$